

SUBHRADEEP SAHA

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SUMMARY

Senior Software Engineer with 4+ years of experience designing, scaling, and operating high-availability Java systems for security-sensitive Tier-0 environments at Oracle Cloud Infrastructure (OCI). Proven track record of leading end-to-end distributed systems initiatives from architectural design through multi-region production deployment and disaster recovery. Skilled in microservices, concurrency, infrastructure automation (Terraform), and AI-assisted engineering workflows, with strong cross-functional collaboration across architecture, security, and partner service teams.

TECHNICAL SKILLS

- **Programming Languages and CS Fundamentals:** Java, C/C++, Python, Data Structures and Algorithms, Object-Oriented Programming(OOPs)
- **Backend, APIs and Distributed Systems:** REST APIs, JAX-RS/Jersey, Microservices, Dropwizard, Google Guice, Jackson, JUnit/Mockito, Swagger, Concurrency
- **Cloud and Infrastructure as Code:** OCI, Terraform, Docker
- **Security, Access Control and Databases:** OCI Vault, Cryptography, Autonomous Database, SQL/NoSQL
- **Developer Tools and AI-Assisted Engineering:** Git, Bitbucket, GitHub, Maven, IntelliJ IDEA, Jira, Confluence, OpenAI Codex
- **CI/CD and Observability:** CI/CD, TeamCity, Grafana

PROFESSIONAL EXPERIENCE

Senior Member of Technical Staff (IC3), Oracle

Oct 2025 - Present

Oracle Stability Initiative

- Leading the architectural design, implementation, and rollout of an enhanced cross-region backup model for OCI Secrets Management's internal Tier-0 data-plane service, improving DR reliability and recovery scalability under Oracle's Stability Initiative.
- Implemented the application and infrastructure transition for critical backup and replication workflows, enabling automatic support for newly launched regions without additional setup; deployed across 100% of OCI's largest commercial realm with rollout in progress for government and national security regions.

Member of Technical Staff (IC2), Oracle

Jun 2022 - Sep 2025

Secrets Management team - a Tier 0 service within the Identity and Security organization of Oracle Cloud Infrastructure (OCI)

Service Disaster Recovery

- Led the end-to-end design and production rollout of Disaster Recovery for OCI's internal tier-0 secrets management service across 100% of production regions, covering bare-metal and VM deployments and strengthening resilience against power outages, data loss, and service disruptions.
- Authored high-level and low-level architecture and design documentation and drove security, privacy, compliance, and engineering reviews, establishing a reference implementation adopted across other Oracle Tier-0 services.

OCI Auto Secret Rotation

- Delivered the global feature launch by building Java validation and continuous-monitoring workflows for real-time failure detection, reducing failure detection time by approximately 90%.
- Automated prerequisite infrastructure with Terraform across Autonomous Database, OCI Functions, Virtual Cloud Network (VCN), routing tables, and related dependencies.

Integration with OCI Network Based Access Control (NBAC)

- Designed and implemented an end-to-end integration with OCI's Network Based Access Control (NBAC), eliminating approximately 100% of redundant READ API calls and improving API performance, reliability, and access security.

Service Authorization

- Added a comprehensive suite of positive and negative authorization tests for all service APIs, thereby strengthening the overall security posture by ~75%.

Immediate Secret Deletion

- Developed a robust whitelisting mechanism for critical customers, enabling them to swiftly delete secrets in response to business requirements and emergency scenarios, improving operational efficiency and safety.

Project Intern, Oracle

Jun 2021 - Jul 2021

- Developed an in-house browser extension backed by Oracle Vault Service for credential management, reducing reliance on third-party tools while preserving security and operational integrity.

EDUCATION

Bachelor of Technology in Computer Science and Engineering, National Institute of Technology (NIT) Trichy

Aug 2018 - May 2022

KEY ACHIEVEMENTS

- **Oracle Org-wide Quarterly Spot Award:** Awarded for leading the Service Disaster Recovery initiative.
- **Oracle Integrity Hero Nomination:** Nominated for uncovering significant performance issues in OCI NBAC integration.
- **HackNITR 3.0 Winner:** Best use of Auth0 in HackNITR hackathon. [[Submission Link](#)]
- **Top Contributor, NJACK Winter of Code 2020:** Open-source program organized by CS Club, IIT Patna. [[Post Link](#)]